

Calibration of Microscopic Penalty Parameters in the Combined Finite–Discrete-Element Method

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Abstract: With unique advantages in simulating solid fracture, the combined finite–discrete-element method has become increasingly popular in the rock mechanics field. Nevertheless, a large number of microscopic parameters in the method need to be specified and many of them cannot be directly measured via laboratory tests. Therefore, these parameters must be determined via a calibration procedure. To calibrate appropriate microscopic parameters, we should find the relationship between them and the macroscopic mechanical response. In these microscopic parameters, the normal and tangential penalty parameters play a very important role. Thus, in this paper we study the effect of penalty parameters on macroscopic mechanical parameters in the combined finite–discrete-element method. It is found that as the penalty parameters increase, the elastic modulus and Poisson's ratio gradually approach the experimental values. However, the further increase of penalty parameters has little effect on the elastic modulus and Poisson's ratio when they reach 100 times the elastic modulus. Based on this conclusion, a simple method for the calibration of penalty parameters is proposed, which can be used to quickly determine the penalty parameters according to the uniaxial compression test. DOI: 10.1061/(ASCE)GM.1943-5622.0001686. © 2020 American Society of Civil Engineers.

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Introduction

Continuum methods [FEM, boundary-element method (BEM)] and discontinuum methods [discrete-element method (DEM)] (Cundall 1971; Cundall 1988; Cundall and Strack 1979), [discontinuous deformation analysis (DDA)] (Shi 1992a, Yang et al. 2018), [numerical manifold method (NMM)] (Shi 1992b, Wu et al. 2019; Yang et al. 2016) have been commonly used in studies of rock deformation and fracturing behavior (Cheng et al. 2017; Dorren 2003; Jing 2003; Tang et al. 2019). However, although the continuum methods can well simulate the deformation, their continuity assumption makes it difficult for those methods to handle the contact and the complex crack initiation and propagation (Baoquan and Tannant 2007; Cundall and Strack 1979). Meanwhile, discontinuum methods such as the DEM have limitations in dealing with deformation or crack initiation and propagation (Lisjak and Grasselli 2010; Williams and John 1988; Williams and Mustoe 1987).

Therefore, combining the advantages of the two different methods (Ghaboussi 1988), Munjiza et al. (Munjiza 2004; Munjiza et al. 1999b) proposed the combined finite–discrete-element method (FDEM). It uses triangular finite-element meshes for discretization, which makes the pre-processing very simple, and the initial model has no gaps compared with the particle DEM. Meanwhile, a joint element with zero initial thickness is inserted between the adjacent

triangular elements. The crack initiation and propagation can be simulated by the breaking of the joint element. Thus, the characterization of the crack is very intuitive. In addition, the FDEM also well preserves the concepts of stress and strain in continuum mechanics.

Based on these superiorities, the FDEM is most suitable for simulating rock fracturing (Lei et al. 2015; Lei et al. 2019; Lisjak and Grasselli 2014; Liu and Deng 2019; Mahabadi et al. 2014; Mahabadi et al. 2012; Munjiza and Andrews 1998; Munjiza et al. 1999b; Rougier et al. 2014; Yan and Jiao 2018; Yan and Jiao 2019a; Yan and Jiao 2019b; Yan and Zheng 2016; Yan and Zheng 2017a; Yan and Zheng 2017b; Yan and Zheng 2017c; Yan and Zheng 2017d; Yan et al. 2016; Yan et al. 2018; Yan et al. 2019a; Yan et al. 2019b; Yan et al. 2019c). Although this method was put forward as early as 1995 (Munjiza et al. 1995) and a monograph about the FDEM was published in 2004 by Munjiza (2004), the relevant researches and applications were still rare before 2010. After that, it became increasingly popular in the field of rock mechanics, especially in rock fracture modeling (Fukuda et al. 2019; Gao et al. 2019; Li et al. 2019; Vazaios et al. 2019). However, the FDEM needs plenty of microscopic parameters and many of them are difficult to determine; for example, a total of 13 input parameters for the joint element (cohesive c , internal friction angle ϕ , energy release rate I G_{JI} , energy release rate II G_{JII} , tensile strength T_s , normal penalty parameter pf_n , and tangential penalty parameter pf_s) and the triangular element (elastic modulus E , Poisson's ratio ν , damping coefficient μ , density ρ , normal penalty parameter p_n , and tangential penalty parameter p_t). Some microscopic parameters cannot be directly obtained from basic rock mechanics tests. Meanwhile, the relationship between microscopic parameters and macroscopic mechanical parameters is still unclear, which both result in difficulty using the FDEM in practical engineering. Although Tatone and Grasselli (2015) presented a calibration method for determining microscopic parameters in the FDEM, the calculated amount of the calibration method is too large. Therefore, this paper attempts to explore the issue in depth.

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Although the FDEM absorbs the advantages of the FEM and DEM, it is a DEM in nature. In the conventional DEM (Cundall 1971; Cundall and Strack 1979), the microscopic parameters need to be calibrated (Zhou et al. 2017) and the determination of stiffness parameters is particularly important. For example, Kazerani and Zhao (2010) studied the effect of stiffness parameters on the elastic modulus and Poisson's ratio. Similarly, in the FDEM, there exist some microscopic parameters, called penalty parameters, that correspond to the stiffness parameters in the conventional DEM. However, the unit of the penalty parameter (Pa) in the FDEM is different from the unit of the stiffness parameter (N/m) in the DEM, which means the conclusions about the stiffness parameters in the DEM cannot be directly applicable to the FDEM. Therefore, we need to specifically study the effects of penalty parameters on macroscopic mechanical properties and simplify the penalty parameter calibration process in the FDEM.

This paper is divided into the following parts. Firstly, we briefly introduce the basic principles of the FDEM, including the constitutive model of the joint element, the system governing equation, and the contact force between the triangular elements. Then, we investigate the effect of penalty parameters on macroscopic mechanical parameters and put forward a calibration method for determining the penalty parameters. Finally, the calibration method is used to determine the penalty parameters of numerical samples of Jinping T2b marble (Zhang et al. 2010). In addition, the calibrated macroscopic mechanical parameters are compared with the experimental values.

Fundamentals of the FDEM

Constitutive Model of Joint Elements

The basic idea of the FDEM is to divide the continuum into triangular elements and insert the joint element with zero initial thickness between the adjacent triangular elements, as shown in Fig. 1. Then, crack initiation and propagation are simulated by the breaking of the joint element. A detailed introduction to the FDEM can be found in the book *The Combined Finite-Discrete-element Method* (Munjiza 2004). In this paper, we just focus on the basic theory related to the fracture constitutive and penalty parameters.

System Governing Equation of the FDEM

Eq. (1) presents the system governing equation of the FDEM. Like the DEM, the governing equation of the node is solved by the Euler

method (Munjiza et al. 2004)

$$\mathbf{M}\ddot{\mathbf{x}} + \mathbf{C}\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}) \quad (1)$$

where $\mathbf{M}$ = the lumped mass matrix; $\mathbf{C}$ = the damping matrix; and $\mathbf{F}(\mathbf{x})$ = the total nodal force vector, which comprises the nodal force $\mathbf{F}_c$ caused by the contact force, the nodal force $\mathbf{F}_d$ caused by the deformation of the triangular element, the nodal force $\mathbf{F}_j$ caused by the deformation of the joint element, and nodal force $\mathbf{F}_e$ caused by the external load. The damping matrix $\mathbf{C}$ is used to solve the static problem with the dynamic relaxation, which is given by

$$\mathbf{C} = \mu \mathbf{I} \quad (2)$$

where μ = the damping coefficient; and $\mathbf{I}$ = the unit matrix. According to the single-degree-of-freedom mass spring system, the critical damping coefficient is given by (Munjiza 2004)

$$\mu = 2h\sqrt{\rho E} \quad (3)$$

where h = element size; ρ = density; and E = elastic modulus. The critical damping coefficient can be used to consume the kinetic energy of the system at the fastest speed.

Contact Force between Triangular Elements

The FDEM converts the contact force between blocks into that between triangular elements. In order to prevent the two triangular elements in contact from being embedded with each other, the contact forces, including the normal and the tangential contact forces, with equal magnitude but opposite direction, are respectively applied to the two triangular elements. Beyond that, the two forces involve many important microscopic parameters, which will be introduced in the subsequent section. Next, we introduce the calculation of contact force briefly.

Normal Contact Force

As shown in Fig. 2, the two triangular elements in contact are referred to as contactor triangular element β_c and target triangular element β_t . The normal contact force for the contactor triangular element embedded in the target triangular element with the area dA is given by (Munjiza 2004; Munjiza and Andrews 2000)

$$d\mathbf{f}_{cl} = -p_n \mathbf{grad} \varphi_t(P_t) dA \quad (4)$$

where P_t = a point located in the target triangular element β_t inside the overlapping area; $\varphi_t(P_t)$ = the potential of point P_t in the target triangular element β_t ; $\mathbf{grad}$ = gradient; and p_n = normal penalty parameter.

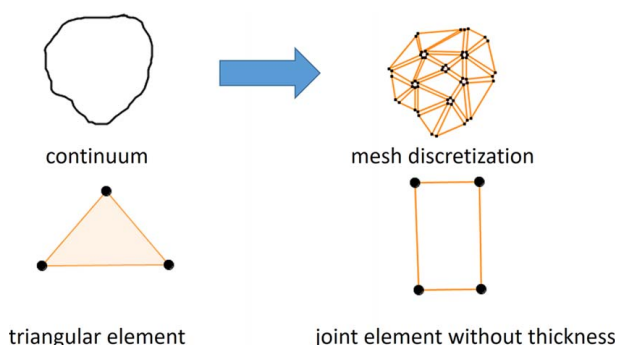


Fig. 1. Meshing of a continuum in the FDEM.

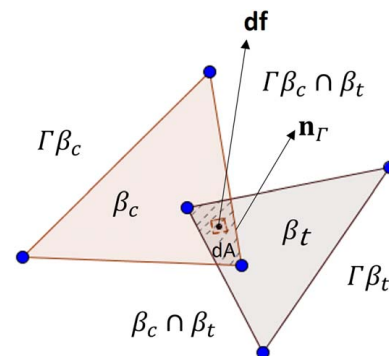


Fig. 2. Schematic diagram of contact force between triangular elements.

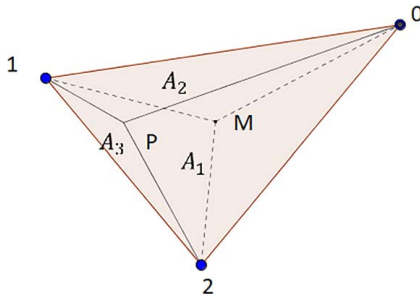


Fig. 3. Potential of point P in a triangular element.

Similarly, the normal contact force for the target triangular element β_t embedded in the contactor triangular element β_c with the area dA is given by

$$d\mathbf{f}_t = -p_n \mathbf{grad} \varphi_c(P_c) dA \quad (5)$$

At the same time, the contactor triangular element β_c is also subjected to the reaction forces with equal magnitude but opposite direction

$$d\mathbf{f}_{c2} = p_n \mathbf{grad} \varphi_c(P_c) dA \quad (6)$$

Hence, the total normal contact force of the overlapping area dA (the target triangular element on the contactor triangular element) is given by

$$d\mathbf{f}_c = d\mathbf{f}_{c1} + d\mathbf{f}_{c2} \quad (7)$$

According to Eqs. (4), (6), and (7) can be rewritten as

$$d\mathbf{f}_c = p_n [\mathbf{grad} \varphi_c(P_c) - \mathbf{grad} \varphi_t(P_t)] dA \quad (8)$$

By integrating Eq. (8), the total normal contact force of the entire overlapping area can be obtained (Munjiza 2004)

$$\mathbf{f}_c = p_n \int_{\Gamma_{\beta_t \cap \beta_c}} [\mathbf{grad} \varphi_c(P_c) - \mathbf{grad} \varphi_t(P_t)] dA \quad (9)$$

According to the Green formula, the surface integral in Eq. (9) can be converted to the line integral over the boundary of the overlapping area

$$\mathbf{f}_c = p_n \int_{\Gamma_{\beta_t \cap \beta_c}} \mathbf{n}_\Gamma (\varphi_c - \varphi_t) d\Gamma \quad (10)$$

where $\Gamma_{\beta_t \cap \beta_c}$ = the boundary of the overlapping area of two triangular elements; and $\mathbf{n}_\Gamma$ = the outward normal unit vector of the boundary of the overlapping area.

The potential function φ should be constant at the boundary of the discrete triangular element (Munjiza 2004). In this way, the total work done by the normal contact force is zero, which is the only requirement for the definition of the potential function. According to this, Munjiza (2004) defined the following potential function

$$\varphi(P) = \min \left\{ \frac{3A_1}{A}, \frac{3A_2}{A}, \frac{3A_3}{A} \right\} \quad (11)$$

where A_i ($i = 1, 2, 3$) = area of the sub-triangle constituted by point P and the three edges of the triangular element, as shown in Fig. 3.

As illustrated in Fig. 3, the triangular element is divided into three sub-triangles by the center point M and the three edges of the triangular element. According to Eq. (11), the potential is equal to 1 at the center of the triangular element and is equal to 0 at the boundary. For point P in sub-triangle $M12$, the potential at

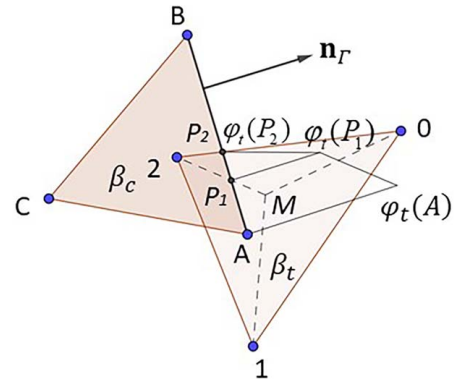


Fig. 4. Normal contact force of the target triangular element on the edge AB of the contactor triangular element.

the point can be further rewritten as

$$\varphi(P) = \frac{3A_3}{A} = \frac{3h_{P-12}}{h_{0-12}} \quad (12)$$

where h_{P-12} = the distance from point P to edge 12; and h_{0-12} = the distance from point 0 to edge 12.

As shown in Fig. 4, points P_1 and P_2 are the intersections of edge AB with midline $M2$ and edge 02 of the target triangular element, respectively. According to Eq. (10), the normal contact force of the target triangular element on edge AB is given by

$$\mathbf{f}_c = p_n \int_{l_{AB}} \mathbf{n}_\Gamma (0 - \varphi_t) dl = -\mathbf{n}_\Gamma p_n \int_{l_{AB}} \varphi_t dl \quad (13)$$

In Fig. 4, φ_t is piecewise linear distributed on edge AB , and Eq. (13) can be rewritten as

$$\mathbf{f}_c = -\mathbf{n}_\Gamma \frac{1}{2} p_n ((\varphi_t(A) + \varphi_t(P_1))l_{AP_1} + (\varphi_t(P_1) + \varphi_t(P_2))l_{P_1P_2}) \quad (14)$$

where $\varphi_t(A)$, $\varphi_t(P_1)$, and $\varphi_t(P_2)$ can be obtained by

$$\varphi_t(A) = \frac{h_{A-12}}{h_{0-12}}, \quad \varphi_t(P_1) = \frac{h_{P_1-12}}{h_{0-12}}, \quad \varphi_t(P_2) = \frac{h_{P_2-20}}{h_{1-20}} = 0 \quad (15)$$

The normal contact force on the other boundaries of the overlapping area can also be obtained according to the above method. The total normal contact force of the triangular element is the vector sum of the normal contact forces on all boundaries.

Tangential Contact Force

It should be noted that there is a tangential contact force at every boundary of the contact overlapping area. Therefore, the total tangential contact force is the vector sum of the tangential contact forces on all boundaries. Here, only the calculation of the tangential contact force on one boundary of the overlapping area is described in detail. For other boundaries, the tangential contact force can be obtained in the same way. Taking the edge AB of the contactor triangular element in Fig. 5 as an example, the calculation of the tangential contact force is introduced.

Using the incremental mode, the Coulomb friction force is given by

$$\mathbf{F}_t^t = \mathbf{F}_t^{t-\Delta t} - p_s \Delta \mathbf{u}_s L \quad (16)$$

where $\mathbf{F}_t^t$ = friction force at the current time step; $\mathbf{F}_t^{t-\Delta t}$ = friction force at the previous time step; p_s = the tangential penalty parameter; L = contact length, i.e., the length of the line segment AP_2 in Fig. 5; and $\Delta \mathbf{u}_s$ is the tangential relative displacement increment in one time step, which can be obtained as follows.

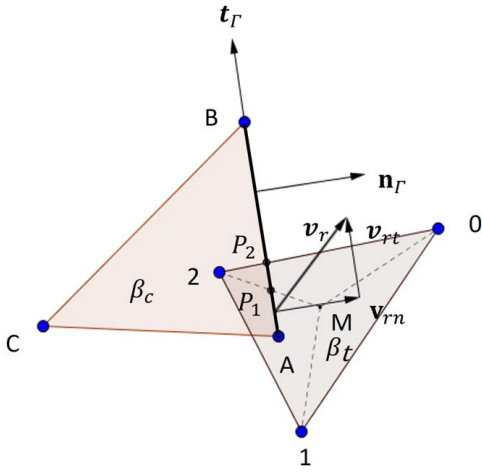


Fig. 5. Tangential contact force of the target triangular element on the edge AB of the contactor triangular element.

First, according to the moment equilibrium, the action point of the normal contact force of the overlapping boundary AP_2 is given by

$$\mathbf{x}_F = \mathbf{x}_A + \lambda_F \mathbf{x}_{P_2} \quad (17)$$

where $\mathbf{x}_A$, $\mathbf{x}_{P_2}$ = coordinate vector of points A and P_2 , respectively; and λ_F is given by

$$\lambda_F = \frac{\lambda_1 \lambda_2 \varphi_1 + \lambda_2^2 \varphi_1 + \lambda_1^2 \varphi_A}{3(\lambda_2 \varphi_2 + \lambda_1 \varphi_A)} \quad (18)$$

where λ_1 and λ_2 are calculated by

$$\lambda_1 = \frac{|\overrightarrow{AP_1}|}{|\overrightarrow{AB}|} \quad (19)$$

$$\lambda_2 = \frac{|\overrightarrow{AP_2}|}{|\overrightarrow{AB}|} \quad (20)$$

Then the relative velocity of the contactor triangular element relative to the target triangular element at the action point $\mathbf{x}_F$ of the normal contact force is given by

$$\mathbf{v}_r = \mathbf{v}_c(\mathbf{x}_F) - \mathbf{v}_t(\mathbf{x}_F) \quad (21)$$

where $\mathbf{v}_c(\mathbf{x}_F)$ and $\mathbf{v}_t(\mathbf{x}_F)$ = velocities of the contactor triangular element and the target triangular element at action point $\mathbf{x}_F$ of the normal contact force of edge AB , respectively.

As demonstrated in Fig. 5, the velocity $\mathbf{v}_c(\mathbf{x}_F)$ of the contactor triangular element and the velocity $\mathbf{v}_t(\mathbf{x}_F)$ of the target triangular element at the action point can be obtained through the interpolation according to the velocity of the vertex of the triangular elements

$$\left. \begin{aligned} \mathbf{v}_c(\mathbf{x}_F) &= (1 - \lambda_F) \mathbf{v}_A + \lambda_F \mathbf{v}_B \\ \mathbf{v}_t(\mathbf{x}_F) &= N_0 \mathbf{v}_0 + N_1 \mathbf{v}_1 + N_2 \mathbf{v}_2 \end{aligned} \right\} \quad (22)$$

where N_0 , N_1 , N_2 = the interpolation shape function of the action point of the normal contact force in the target triangular element.

The normal relative velocity $\mathbf{v}_{rn}$ at the action point is given by

$$\mathbf{v}_{rn}(\mathbf{x}_F) = (\mathbf{v}_r(\mathbf{x}_F) \cdot \mathbf{n}_r) \mathbf{n}_r \quad (23)$$

The tangential relative velocity $\mathbf{v}_{rt}$ at the action point is given by

$$\mathbf{v}_{rt}(\mathbf{x}_F) = \mathbf{v}_r(\mathbf{x}_F) - \mathbf{v}_{rn}(\mathbf{x}_F) \quad (24)$$

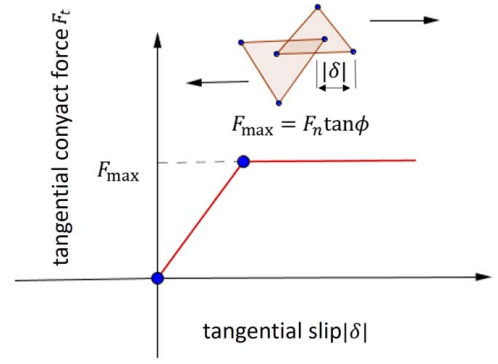


Fig. 6. Relationship of tangential slip amount and tangential contact force for one contact pair.

The tangential relative displacement increment in one time step is given by

$$\Delta \mathbf{u}_s = \mathbf{v}_{rt}(\mathbf{x}_F) \Delta t \quad (25)$$

As exhibited in Fig. 6, if the tangential contact force calculated by Eq. (16) satisfies $|\mathbf{F}'_t| \geq F_n \tan \phi$, then it is finally given by

$$\mathbf{F}_t = \mathbf{F}'_t \quad (26)$$

If the tangential contact force calculated by Eq. (16) satisfies $|\mathbf{F}'_t| \leq F_n \tan \phi$, then it is finally given by

$$\mathbf{F}_t = \frac{\mathbf{F}'_t}{|\mathbf{F}'_t|} u F_n \quad (27)$$

At the same time, the tangential slip amount is set as

$$u_s = \frac{|\mathbf{F}_t|}{p_s h} \quad (28)$$

Thus, the tangential frictional force of the edge AB is obtained according to Eqs. (26)–(28), where Eq. (28) is used to ensure that the maximum tangential contact force is $u F_n$ in the subsequent time steps.

Distribution of Contact Force

In this section, we describe how to distribute the contact force to the nodes of the triangular element.

Still taking the edge AB in Fig. 5 as an example, the total contact force of the edge AB can be given by

$$\mathbf{F}_c = \mathbf{F}_n + \mathbf{F}_t \quad (29)$$

According to the action point of normal contact force $\mathbf{x}_F$ obtained in Eq. (17), the contact forces assigned to the contactor and target triangular elements in Fig. 7 are calculated by

$$\left. \begin{aligned} \mathbf{F}_{NA} &= (1 - \lambda_F) \mathbf{F}_c \\ \mathbf{F}_{NB} &= \lambda_F \mathbf{F}_c \\ \mathbf{F}_{NC} &= 0 \end{aligned} \right\} \quad (30)$$

$$\left. \begin{aligned} \mathbf{F}_{N0} &= N_0 \mathbf{F}_c \\ \mathbf{F}_{N1} &= N_1 \mathbf{F}_c \\ \mathbf{F}_{N2} &= N_2 \mathbf{F}_c \end{aligned} \right\} \quad (31)$$

Fracture Constitutive of the Joint Element in the FDEM

In the FDEM, crack initiation and propagation are expressed by the breaking of the joint element. Therefore, the fracture

criterion of the joint element is critical to fracture simulation in the FDEM.

As shown in Fig. 8(a), the normal opening amount o of the two edges connected by the joint element increases until reaching critical value o_p ; normal bond stress σ also increases and reaches tensile strength f_t of the joint element. Then, normal bond stress σ gradually decreases to zero until normal opening amount o reaches maximum value o_r . At this moment, the joint element breaks and a tensile crack is generated.

As shown in Fig. 8(b), if the tangential slip amount s of the two edges connected by the joint element increases until reaching critical value s_p , tangential bond stress τ also increases and reaches shear strength f_s of the joint element. Then, tangential bond stress τ gradually decreases to zero until tangential slip s reaches maximum value s_r . At this moment, the joint element breaks and a shear crack is generated.

In addition, another state (i.e., Mode I-II fracture type, tensile-shear mixing fracture) is that normal opening and tangential sliding both occur for the two edges of the joint element. The normal opening amount and tangential slip amount satisfy the following condition:

$$\left(\frac{o - o_p}{o_r - o_p}\right)^2 + \left(\frac{s - s_p}{s_r - s_p}\right)^2 \geq 1 \quad (32)$$

Maximum normal opening amount o_r and maximum tangential slip amount s_r are related to tensile strength f_t , shear strength f_s , energy release rate Gf_I of Mode I fracture, and energy release rate Gf_{II} of Mode II fracture. As shown in Fig. 8(c), energy release rates Gf_I and Gf_{II} are expressed by (Lisjak et al. 2014; Munjiza et al. 1999a)

$$Gf_I = \int_{o_p}^{o_r} \sigma(o) do \quad (33)$$

$$Gf_{II} = \int_{s_p}^{s_r} \tau(s) ds \quad (34)$$

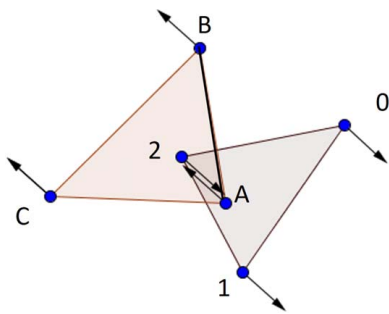


Fig. 7. Distribution of contact force for the contact pair of the triangular element.

Sensitivity Analysis of the Penalty Parameter in the FDEM

As we know, the microscopic parameters for the DEM only can be determined by matching the macroscopic mechanical response of numerical experiments with basic rock mechanics tests (Potyondy and Cundall 2004). Similarly, the FDEM also has some microscopic parameters that can be divided into three components. The first part is the microscopic parameters of the triangular elements, including elastic modulus E_p , Poisson's ratio ν_p , and density ρ . The second part is the microscopic parameters of joint elements, including tensile strength f_t , cohesion c , internal friction angle ϕ , and fracture energy release rates Gf_I and Gf_{II} (Potyondy and Cundall 2004). The third part is the microscopic penalty parameters, including normal penalty parameter p_n and tangential penalty parameter p_s of the triangular element and penalty parameter p_{fn} and tangential penalty parameter p_{fs} of the joint element. Among them, only the first part can take the parameters of actual rock as Table 1 shows; the other two parts are not directly available from basic rock mechanics tests.

Although Tatone and Grasselli (2015) proposed a calibration method for microscopic parameters of the FDEM, the calibration process is extremely intricate and only the microscopic parameters in the second part are mainly studied. Furthermore, in the third part, the effect of penalty parameters on macroscopic mechanical properties is still vague. In order to understand these effects, this paper focuses on microscopic parameters in the third part, i.e., the effect of penalty parameters on macroscopic mechanical properties. Meanwhile, the calibration method for determining penalty parameters is proposed. We perform a sensitivity analysis of penalty parameters in detail and establish a simple and easy-to-use calibration method for determining penalty parameters.

The FDEM requires up to 13 input parameters, none of which can be directly obtained from the basic rock mechanical test except for density ρ . However, E_p and ν_p can be directly taken as the macroscopic elastic modulus E and Poisson's ratio ν of the

Table 1. Input parameters in the FDEM

Input parameter	
Triangular element	Joint element
Density ρ (kg/m ³)	Tensile strength f_t (MPa)
Elastic modulus E_p (GPa)	Cohesive c (MPa)
Poisson's ratio ν_p	Internal friction angle ϕ_f (°)
Damping coefficient μ (kg/m · s)	Energy release rate of I Gf_I (J/m ²)
Internal friction angle ϕ (°)	Energy release of II Gf_{II} (J/m ²)
Normal penalty parameter p_n (GPa)	Joint normal penalty parameter p_{fn} (GPa)
Tangential penalty parameter p_t (GPa)	Joint tangential penalty parameter p_{fs} (GPa)

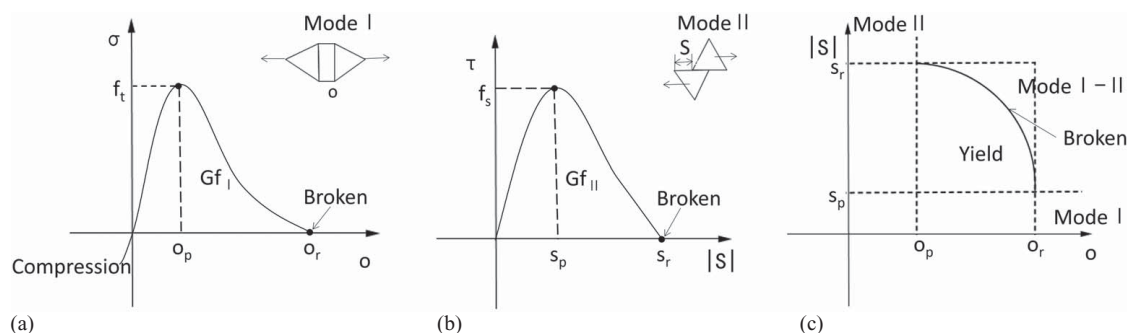


Fig. 8. Fracture criterion of the joint element in the FDEM: (a) tensile failure; (b) shear failure; and (c) tensile shear mixed failure.

material to simplify the calibration process. The damping coefficient can be directly determined according to Eq. (3). In addition, we assume that p_n is equal to p_{fn} and p_s is equal to p_{fs} . This processing strategy is like the stiffness parameters of the bonded-particle model in the DEM. In this way, we only need to carry out a sensitivity analysis for two parameters p_n and p_s . Then, we analyze the effect of these two penalty parameters on the macroscopic parameters.

According to Eqs. (4)–(10) and (16), p_n is used to prevent excessive embedding of the triangular element by applying normal force when embedding occurs; p_s is used to calculate tangential friction when there is relative sliding between triangular elements in contact. Therefore, p_n and p_s have a great effect on the mechanical behavior of numerical samples. In this paper, p_n and p_s will be determined according to the uniaxial compression and Brazilian disc test.

According to the experiment in the literature (Zhang et al. 2010), laboratory parameters of the Baishan Group Jinping marble are as presented in Table 2. The aim of the subsequent calibration of penalty parameters is to make the macroscopic mechanical parameters of the numerical sample close to those listed in Table 2.

Fig. 9 shows the mesh of the numerical samples. The uniaxial compression test sample has a width of 50 mm and a height of 100 mm. The Brazilian disc test sample is 50 mm in diameter. Both samples are divided into the triangular elements with an edge length of 0.8 mm, which is less than 1 mm and meets the requirement of mesh size sensitivity (Tatone and Grasselli 2015). The element size retains the same value for all numerical simulations. Two loading plates are applied at a constant velocity of 0.05 m/s along the opposite direction. Although this loading velocity is much larger than the actual loading velocity, the trial calculation shows that the mechanical response of the system does not change when the loading velocity is less than or equal to this (Tatone and Grasselli 2015).

Table 2. Mechanical parameters of specimens of Jinping T2b marble

Parameter	Values
Density ρ (kg/m ³)	2,700
Elastic modulus E (GPa)	43.73
Poisson's ratio ν	0.19
Tensile strength σ_t (MPa)	13
Uniaxial compressive strength σ_c (GPa)	113.36
Cohesive c (MPa)	37.28
Internal friction angle ϕ (°)	33.69

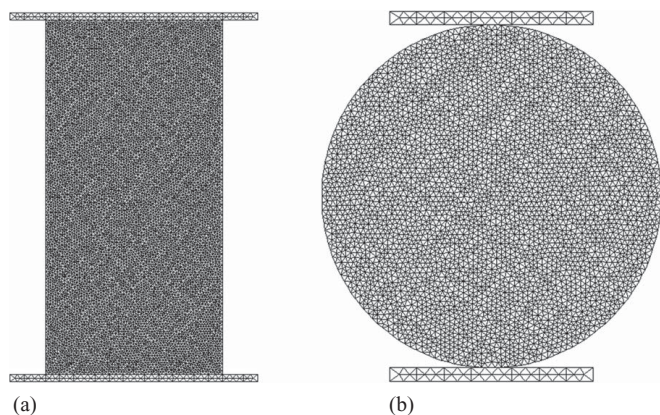


Fig. 9. Numerical model for calibration: (a) uniaxial compression; and (b) Brazilian disc.

Normal Penalty Parameter and Tangential Penalty Parameter

In the bond particle model (BPM) in particle flow code (PFC), Potyondy and Cundall (2004) pointed out that the macroscopic elastic modulus of a numerical sample mainly depends on the normal stiffness and tangential stiffness of the contact. Kazerani and Zhao (2010), Gao and Stead (2014), and Ghazvinian et al. (2014) also gave the same conclusion based on the grain-based model (GBM) in Universal Distinct Element Code (UDEC) and three-dimension distinct element code (3DEC). Mahabadi (2012) and Tatone and Grasselli (2015) pointed out that the elastic modulus increases with the increase of the normal and tangential penalty parameters. Similarly, in the FDEM, the macroscopic mechanical parameters of numerical samples are also directly determined by normal and tangential penalty parameters.

As shown in Fig. 10, when p_s/p_n is constant, the macroscopic elastic modulus E and the Poisson's ratio ν of numerical samples increase gradually with the increase of the normal penalty parameter. At the same time, the elastic modulus E and Poisson's ratio ν become ever-closer to those of the laboratory parameters and almost equal to them as penalty parameters go up to 100 E . Then, with the continuing increase of penalty parameters, the variable quantities of the elastic modulus and Poisson's ratio of the numerical model are tiny.

As illustrated in Fig. 11, when p_s/p_n is fixed, the uniaxial compression strength remains substantially constant, but the Brazilian disc tensile strength decreases and eventually stabilizes with the increase of p_n and p_s , as shown in Fig. 12. When p_n and p_s increase to 4,000 GPa (about 100 times the elastic modulus), the tensile strength basically remains unchanged if penalty parameters continue to increase. Inversely, when p_n and p_s are too small, tensile strength increases sharply. That is, tensile strength is not sensitive to p_n and p_s if p_n and p_s are large (usually 100 times or more the elastic modulus E). However, when p_n and p_s are small, the sensitivity of Brazilian disc tensile strength to penalty parameters rises sharply.

Ratio of the Tangential Penalty Parameter to the Normal Penalty Parameter (p_s/p_n)

Based on the BPM model in PFC, Potyondy and Cundall (2004) indicated that the Poisson's ratio is determined by the ratio of normal stiffness to tangential stiffness. Similar conclusions are also indicated by Kazerani and Zhao (2010) and Ghazvinian et al. (2014)

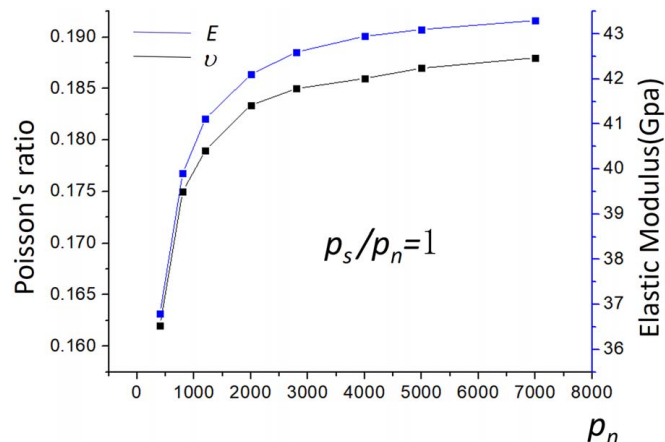


Fig. 10. Effect of the penalty parameter on the macroscopic elastic modulus and Poisson's ratio with constant p_s/p_n .

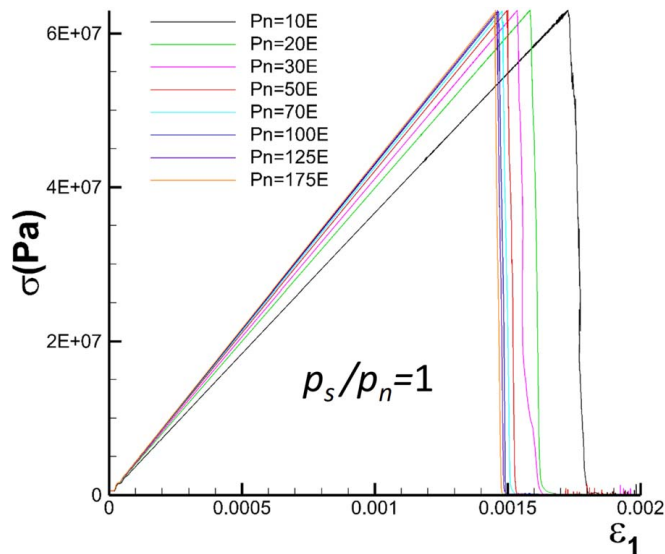


Fig. 11. Effect of the penalty parameter on uniaxial compression strength with constant p_s/p_n .

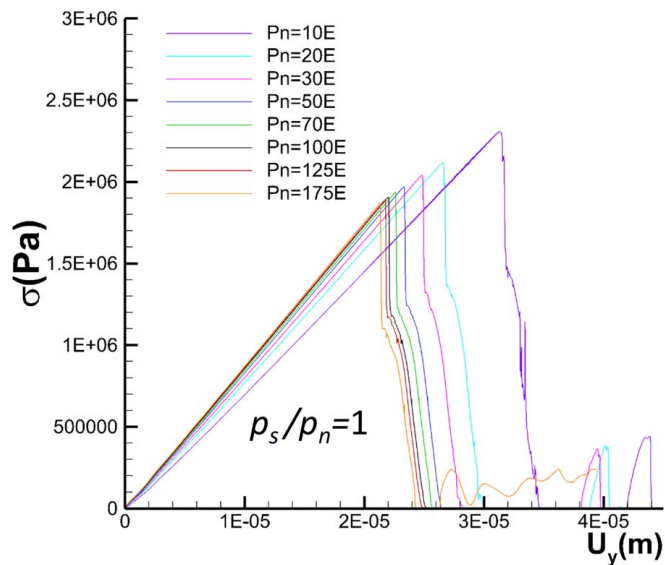


Fig. 12. Effect of the penalty parameter on Brazilian disc tensile strength with constant p_s/p_n .

based on the GBM model in UDEC and 3DEC, respectively. To this end, we study whether there are similar features in the FDEM. As exhibited in Fig. 13, when p_n remains unchanged and p_s/p_n is less than 1, the macroscopic elastic modulus E gradually increases and tends to a fixed value as p_s/p_n increases. However, Poisson's ratio decreases and tends to a fixed value. Then, if p_s/p_n is larger than 1, the elastic modulus and Poisson's ratio both no longer tend to be fixed but change along with the original trend. Since the increase of p_s means the increase of p_s , the model becomes harder and less prone to lateral deformation. Thus, Poisson's ratio becomes smaller. As a result of the lateral constraint, the axial deformation will also be constrained and the elastic modulus will increase. According to numerical simulation results, the macroscopic elastic modulus and Poisson's ratio of the numerical model are closer to the experimental values and more stable when the value of p_s/p_n is close to 1.

As shown in Fig. 14, the sensitivity analysis is carried out on uniaxial compression stress–strain curves with different p_s/p_n ,

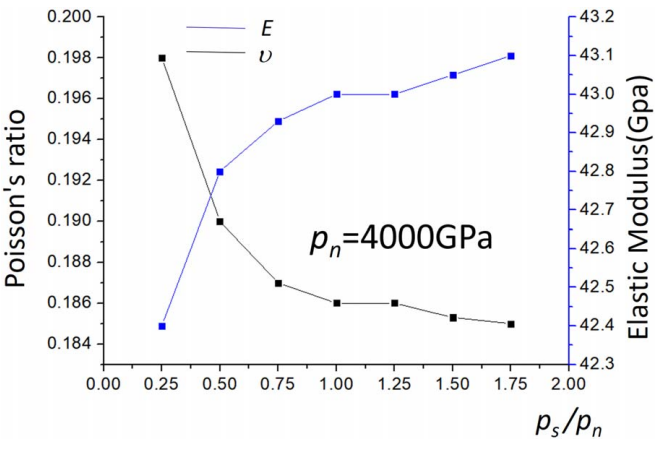


Fig. 13. Effect of p_s/p_n on the macroscopic elastic modulus and Poisson's ratio with $p_n = 4,000$ GPa.

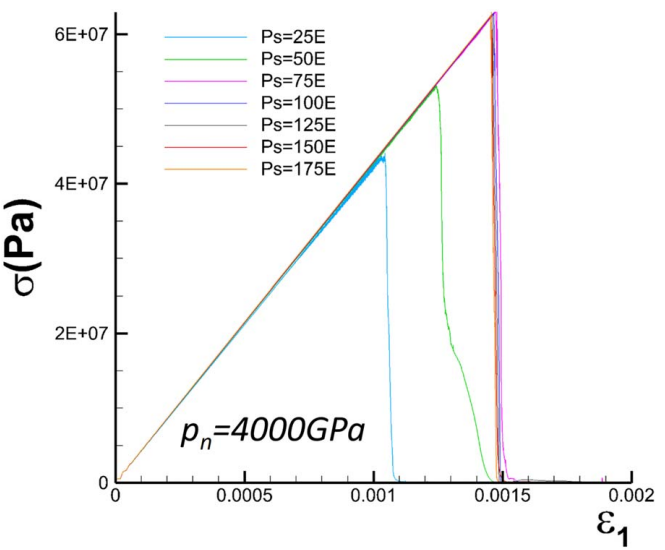


Fig. 14. Uniaxial compression stress–strain curve for different p_s with $p_n = 4,000$ GPa.

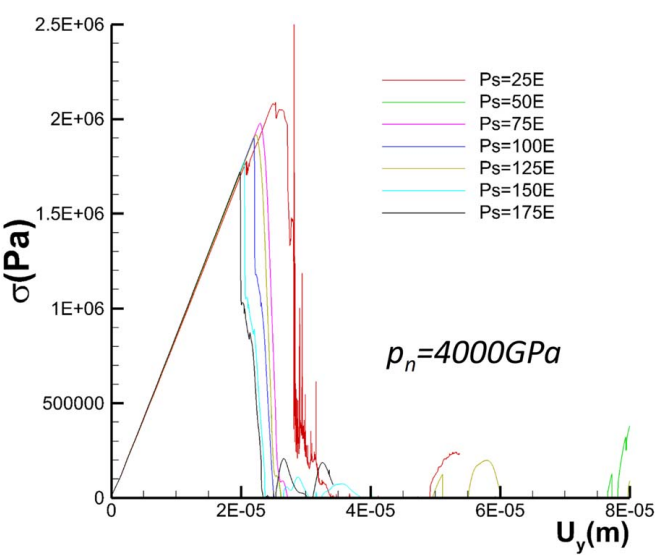


Fig. 15. Tensile stress-vertical displacement curve for different p_s with $p_n = 4,000$ GPa.

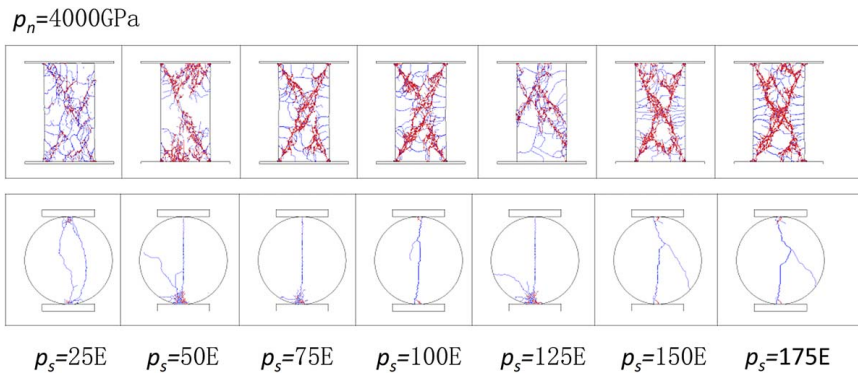


Fig. 16. Failure patterns at post peak of the uniaxial compression and Brazilian disc test for different p_s with $p_n = 4,000$ GPa.

where p_n remains constant at 4,000 GPa and p_s changes. It can be seen that, with the increase of p_s/p_n , uniaxial compression strength increases gradually until p_s reaches 100 E . After that, the peak value changes very little and remains stable basically if p_s continues to increase. In contrast, if p_s is much smaller than 100 E , uniaxial compression strength changes greatly. This phenomenon indicates that different values of p_s have different effects on uniaxial compression strength. This is different from the GBM model. In the GBM model, it is believed that uniaxial compression strength increases all the time with keeping p_n unchanged and p_s/p_n increasing (Ghazvinian et al. 2014). This is a good property of the FDEM compared with GBM model, indicating that if penalty parameters are 100 times or more greater than the elastic modulus and p_s/p_n is larger than 1, they will have very little effect on peak strength. This feature is very favorable for parameter calibration. Therefore, the value and ratio of p_s and p_n control the macroscopic Poisson's ratio and elastic modulus of the numerical sample in the FDEM. However, the effect of penalty parameters on uniaxial compression peak strength is very small when the penalty parameter is more than 100 E and p_s/p_n is larger than 1.

As shown in Fig. 15, when p_n is fixed, tensile strength does not change regularly with the increase of p_s , but it can be seen that the curve shape is better when penalty parameters are larger than 100 E .

As shown in Fig. 16, the failure pattern at the post peak of uniaxial compression and Brazilian disc test gets better when p_s/p_n is closer to 1 with constant p_n . For a uniaxial compression test, if the tangential penalty parameter is less than the normal penalty parameter, the shear crack is more likely to develop laterally and does not pass through the two ends of the numerical sample. That the lateral limit of the numerical sample is less than the axial limit leads to the premature failure. As shown in Fig. 14, uniaxial compression strength decreases significantly when p_s is less than p_n .

Similarly, for $p_s = 4,000$ GPa, with the increase of p_n/p_s , the elastic modulus E and Poisson's ratio ν both increase gradually and tend to be fixed, as shown in Fig. 17.

As illustrated in Fig. 18, keeping $p_s = 4,000$ GPa, uniaxial compression strength gradually stabilizes when p_n is close to 100 E , and the curve shape becomes better when p_s/p_n is around 1. As shown in Fig. 19, with p_n increasing to 100 E , tensile strength gradually tends to be stable, and the curve shape is better when the penalty parameter is above 100 E .

As shown in Fig. 20, when p_s remains unchanged, the failure pattern at the post peak of uniaxial compression and Brazilian disc test is better if p_s/p_n is close to 1. When p_n increases to 7,000 GPa, p_s is relatively small, which leads to non-axial penetration of shear cracks and the reduction of the peak stress. This can also be seen in Fig. 18. However, this situation does not occur

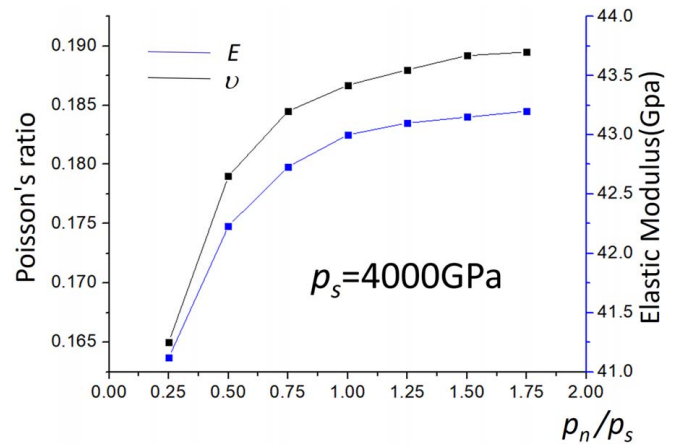


Fig. 17. Effect of p_n/p_s on the macroscopic elastic modulus and Poisson's ratio with $p_s = 4,000$ GPa.

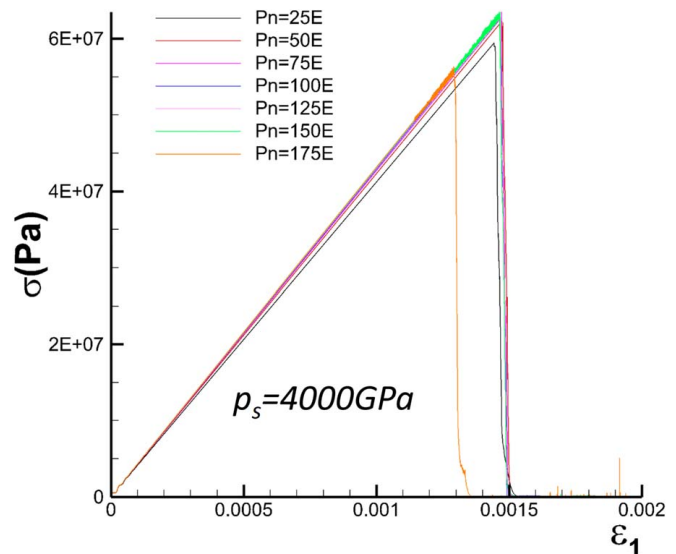


Fig. 18. Uniaxial compression stress-strain curve for different p_n with $p_s = 4,000$ GPa.

when p_n goes down to a very small value. When $p_n < p_s$, even though the shear crack tends to penetrate longitudinally, it can still pass through the sample laterally because the width of the sample is smaller than the length. Therefore, the failure patterns

and peak stress change very little when p_n is decreased. However, as shown in Fig. 18, if p_n is too small (e.g., $p_n = 25 E$), uniaxial compression strength still decreases as a result of the earlier longitudinal penetration of shear cracks. As a result, the decrease of uniaxial compression strength will occur whether p_n is too large or too small, which is reflected in Fig. 18.

In summary, when p_s/p_n is fixed, the macroscopic elastic modulus and Poisson's ratio increase at a faster rate with the increase of p_n and p_s . However, the rate decreases obviously if p_n and p_s are larger than 100 E . Moreover, as both p_s and p_n exceed 100 E , they have no effect on uniaxial compression strength. However, the effects of p_s and p_n on tensile strength have no obvious rule, but the curve shape is better when p_s/p_n is close to 1.

Calibration Method of Penalty Parameters in the FDEM

Based on the above sensitivity analysis of penalty parameters on the macroscopic mechanical response, we propose the following calibration method for penalty parameters in the FDEM.

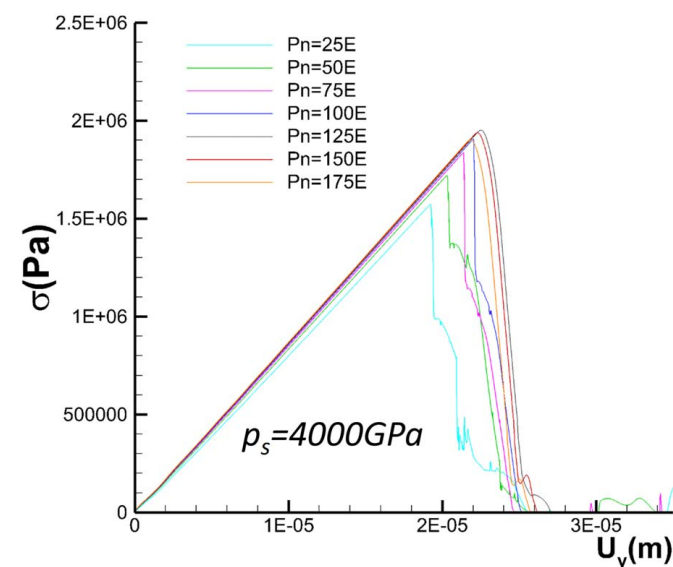


Fig. 19. Tensile stress-vertical displacement curve for different p_n with $p_s = 4,000$ GPa.

At the beginning of the calibration, we need a set of initial parameters, which are directly taken from experimental parameters, i.e., $\rho_p = \rho_{\text{macro}}$, $E_p = E_{\text{macro}}$, $\nu_p = \nu_{\text{macro}}$, $p_n = p_{fn} = E_{\text{macro}}$, $p_f = E_{\text{macro}}$, $c = c_{\text{macro}}$, $\phi = \phi_{\text{macro}}$, $f_t = T_s$ (i.e., tensile strength). Besides, G_{fI} and G_{fII} can be taken larger values to ensure that tensile and shear failure do not occur when the sample is subjected to load. The specific calibration of c , ϕ , f_t , G_{fI} , and G_{fII} can be found in the literature (Potyondy and Cundall 2004). Here ρ_p , E_p , and ν_p are directly obtained from experiments, so they do not need to be calibrated and remain unchanged during subsequent calibration.

1. First, determine the mesh size of the numerical sample. As mentioned above, a mesh size of 1 mm or fewer satisfies the requirements.
2. If p_s/p_n is kept at 1 and penalty parameters are all larger than 100 E , E_{macro} matching the test can be obtained by adjusting p_n . According to the section "Sensitivity Analysis of the Penalty Parameters in the FDEM," under the premise that p_s/p_n is constant, the elastic modulus increases with the increase of p_n . Therefore, p_n can be adjusted by dichotomy until the elastic modulus matches the experimental value.

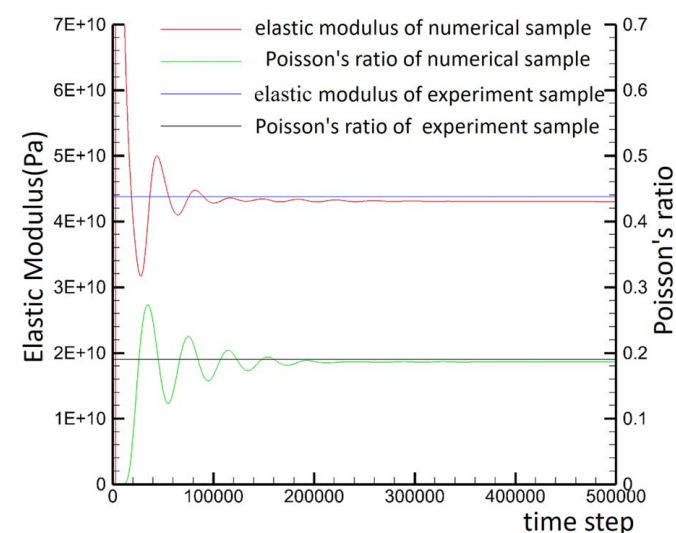


Fig. 21. Comparison of the macroscopic elastic modulus and Poisson's ratio of numerical samples with experimental values of Baishan Group Jinping marble during the uniaxial compression test.

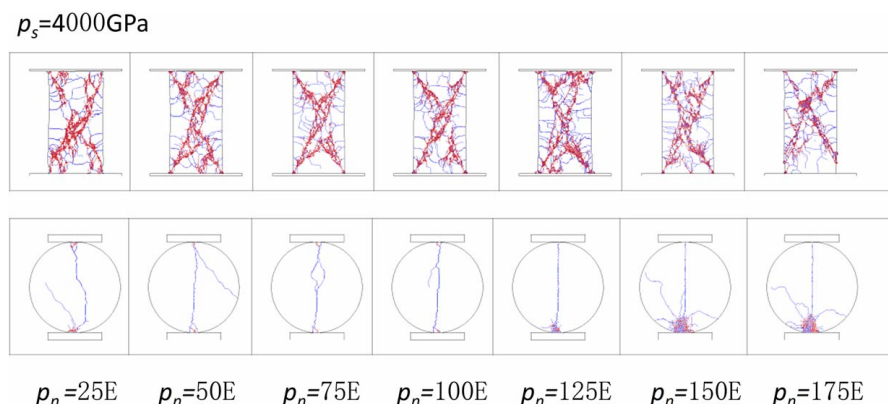


Fig. 20. Failure patterns at post peak of the uniaxial compression and Brazilian disc test for different p_n with $p_s = 4,000$ GPa.

Calibration of Penalty Parameters of Jinping Marble

We use the aforementioned calibration method of penalty parameters to calibrate these parameters of numerical samples according to marble from the Jinping II Baishan Group (Zhang et al. 2010). Taking $p_n = 4,000$ and $p_s = 4,000$ GPa, the calibrated macroscopic deformation parameters of numerical samples are shown in Fig. 21. The macroscopic elastic modulus and Poisson's ratio of the numerical sample are very close to those of the Baishan Group marble, which proves that the calibration method of penalty parameters proposed in this paper is effective. Thus, the macroscopic elastic modulus and Poisson's ratio of the numerical sample matched to the test can be quickly obtained.

Conclusion

In the FDEM, the relationship between the penalty parameter and the macroscopic mechanical parameter is studied. As the normal penalty parameter increases, the macroscopic elastic modulus and Poisson's ratio both increase. However, after the normal penalty parameter goes up to 100 times the elastic modulus, the macroscopic elastic modulus and Poisson's ratio tend to be fixed. Oppositely, with the increase of the tangential penalty parameter, the macroscopic Poisson's ratio decreases and the macroscopic elastic modulus increases.

The normal and tangential penalty parameters have little effect on uniaxial compression strength when they are larger than 100 times the elastic modulus, and their effect on Brazilian tensile strength is not obvious, but the curve shape of the Brazilian disc test is better if normal and tangential penalty parameters are larger than 100 times the elastic modulus and p_s/p_n is close to 1.

As a result of the effect of the aspect ratio on crack propagation, the effect of the tangential penalty parameter on the failure mode and uniaxial compression strength is more obvious than that of the normal penalty parameter.

A simple penalty parameter calibration method is proposed. The penalty parameters can be quickly calibrated according to the basic rock mechanics test.

It should be noted that although only penalty parameters of the numerical sample for Jinping marble are calibrated in this paper, the aforementioned conclusion is also applicable to the numerical sample for other types of rock.

Data Availability Statement

All data, models, and code generated or used during the study appear in the published article.

Acknowledgments

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